

Homework — 1.4.3 Boolean Algebra — ANSWER SHEET

Separate answer sheet / mark scheme — keep for marking.

Mark scheme

Total: Part A 14 + Part B 6 = 20 marks.

Part A

1. [2] Completed match-up (letter F is not used — $X + \neg X = 1$ is the complement identity, not in the list):

Law	Match
Double negation	E
De Morgan's law	A
Distribution	B
Association	D
Commutation	C

Mark: all five correct = 2; two to four correct = 1.

2. [3] $Q = (A + B) \cdot \neg C$:

A	B	C	A+B	$\neg C$	Q
0	0	0	0	1	0
0	0	1	0	0	0
0	1	0	1	1	1
0	1	1	1	0	0
1	0	0	1	1	1
1	0	1	1	0	0
1	1	0	1	1	1
1	1	1	1	0	0

Mark: A+B column correct (1); $\neg C$ column correct (1); Q column correct — $Q = 1$ only when (A OR B) is true AND $C = 0$ (1).

3. [3] The two errors and the correct result:

- **Error 1 (Step 1):** De Morgan's law applied to $\neg(A + \neg B)$ must change the **+** to **·** as well as negating each term — the line should read $Y = \neg A \cdot \neg(\neg B)$, not $\neg A + \neg(\neg B)$ (1).
- **Error 2 (Step 2):** double negation gives $\neg(\neg B) = B$, not $\neg B$ (1).
- **Correct final expression:** $Y = \neg A \cdot B$ (1).

Check (row A=0, B=1): $\neg(0 + \neg 1) = \neg(0 + 0) = 1$ and $\neg A \cdot B = 1 \cdot 1 = 1$ ✓; for A = 1, $\neg(1 + \neg B) = 0$ and $\neg A \cdot B = 0$ ✓.

4. [3] Completed paragraph, in order of the blanks:

1. **XOR** (Sum = $A \oplus B$)
2. **AND** (Carry = $A \cdot B$)
3. **carry-in**
4. **full**
5. **one bit**
6. **clock pulse**

Unused distractors: OR, overflow, one byte.

Mark: 2 correct blanks per mark (6 blanks → 3 marks).

5. (a) [1] Completed Karnaugh map (all eight cells correct):

	BC = 00	BC = 01	BC = 11	BC = 10
A = 0	1	1	0	0
A = 1	1	1	1	0

(b) [2] - Group of **four** covering both rows of the BC = 00 and BC = 01 columns → $\neg B$ (1). - Group of **two** in the A = 1 row covering BC = 01 and BC = 11 → $A \cdot C$, giving the simplified expression $X = \neg B + A \cdot C$ (1).

Groups must be rectangles whose sizes are powers of two; groups may overlap (the cell A=1, BC=01 belongs to both). Verified against the truth table: $\neg B + A \cdot C = 1$ exactly on rows 000, 001, 100, 101, 111.

Part B

6. [3] Symmetric encryption uses the **same key** to encrypt and decrypt, whereas asymmetric encryption uses a **pair of keys** — a **public key** to encrypt and the matching **private key** to decrypt (1 — for the key difference). Symmetric is **faster** but suffers the **key-distribution problem** (the shared key must be transmitted securely) (1). Advantage of asymmetric (any one): the public key can be **shared openly**, which solves the key-distribution problem; it also enables **digital signatures** (sign with private key, verify with public key) (1). (AO1)

7. [3] Round robin gives each process a **fixed time slice (quantum)** of CPU time in turn (1). If a process does not finish within its slice it is **paused and moved to the back of the queue**, and the next process runs; this cycle repeats until all processes complete (1 — it is pre-emptive). Advantage (any one): it is **fair** — every process gets regular CPU time so no process waits indefinitely, and it gives good responsiveness for interactive tasks (1). (AO1)